A Computational View of Medicine

Why AI will matter most where disease becomes a search problem

Birger Moëll and Erica Hughberg
Co-Founders, Eir Space

March 17, 2026

Thesis. The most important medical use of advanced AI is not generic automation. It is high-context reasoning over difficult, patient-specific problems where better synthesis can change diagnosis, treatment choice, escalation, or trial matching. In these settings, disease is better understood as a computational burden distributed across time, modalities, and decision branches than as a static label alone.

Abstract

Artificial intelligence will not improve every part of medicine equally. Its highest-value role is in disease settings where the bottleneck is not data collection or workflow automation but high-context reasoning over multimodal evidence under uncertainty. We argue for a computational view of medicine that treats serious illness as a search problem over longitudinal records, pathology, imaging, genomics, patient-generated data, treatment history, and branching therapeutic options. The key question is not whether a model can produce clinically fluent text, but whether additional inference changes the next real decision for a specific patient. This constraint applies to clinicians and AI systems alike: both face the same search problem, and the quality of the search depends on the depth and structure of reasoning applied. We synthesize evidence from precision oncology, rare-disease genomics, antimicrobial discovery, sepsis surveillance, patient-reported outcomes, open-notes adoption, and N-of-1 medicine. Patient access to records is necessary but insufficient; because clinical reasoning is distributed across specialists and institutions—with the patient as the only persistent entity—data access becomes powerful only when patients and clinicians can repeatedly compute over the full timeline of illness. The result is a framework for directing AI toward cases where reasoning is the bottleneck, outcomes depend on synthesis quality and calibration, and patient-held data can become a force multiplier.

1 Introduction

Medicine is organized around disease labels, organ systems, and specialty workflows. These abstractions are useful but obscure where the real difficulty lies. In the cases that matter most, the limiting factor is neither the absence of a diagnosis code nor a single decisive test. It is the burden of synthesis: reconciling longitudinal evidence, deciding which signals matter, ranking interventions, and updating the model of a case as new data arrive.

Advanced AI should be judged through that lens. The central question is not whether a system can imitate clinical language but whether additional inference changes a meaningful next decision in settings where the decision is difficult, high-stakes, and patient-specific. This extends Topol's case for human-machine medicine by placing computational leverage, rather than generic augmentation, at the center [34].

This is not the first such upgrade. Evidence-based medicine replaced individual recall and apprenticeship-derived judgment with structured synthesis over aggregated trial data. Cochrane reviews are, in effect, pre-computed answers to clinical questions. The computational view is the next step. Where EBM asked "what does the average patient in this trial tell us about this patient?", the computational view asks "what does the full multimodal, longitudinal record of *this* patient tell us about *this* decision?" The question is no longer whether medicine should be computational, but how deep the computation should go and where to direct it.

Clinical reasoning can be carried out by clinicians, AI systems, or human-AI teams. The important question is not who reasons in the abstract, but whether additional reasoning changes the next meaningful decision for a patient. This makes the computational view legible inside ordinary medical work: every encounter already runs on a reasoning budget, and many failures of care are failures of time, memory, synthesis, or search rather than failures of data collection alone.

This constraint is not unique to machines. Physicians employ both rapid pattern recognition and slower deliberate analysis—dual-process reasoning—switching between modes depending on case complexity [6]. But time per patient is finite. The consequences are measurable: approximately 5% of US adults—over 12 million people—experience a diagnostic error in outpatient care each year [31], and an estimated 795,000 Americans suffer permanent disability or death annually from diagnostic errors, concentrated in cardiovascular events, infections, and cancer [24]. If a doctor had twice as long to reason through a complex case, would they reach a better conclusion? For routine cases, no: the protocol is clear and more thought adds nothing.

For high-burden cases—a confusing constellation of symptoms, an ambiguous genomic result, a cancer that has stopped responding—more reasoning time genuinely changes the output. The "diagnostic timeout" in emergency medicine, a structured pause to reconsider the differential, reduces diagnostic error [6, 9]. Primary care consultation lengths range from 5 minutes to over 20 across healthcare systems [13]; if reasoning time is the bottleneck, this variation should predict diagnostic quality.

Reasoning-focused language models can engage in structured clinical reasoning—hypothesis generation, differential weighting, evidence integration—that parallels this deliberate process [21]. On complex diagnostic cases from the *New England Journal of Medicine*, GPT-4 included the correct diagnosis in 64% of cases, compared to 36% for medical-journal readers [18]. Med-PaLM 2, a medically fine-tuned model, achieved 86.5% on USMLE-style questions and was preferred over physician answers on eight of nine clinical axes [32]. These results suggest that sufficiently deep AI reasoning can match or exceed clinician-level synthesis on specific benchmarks. But the relationship between reasoning depth and decision quality is not monotonic. In DeepSeek R1, shorter reasoning chains correlate with accuracy; excessively long chains correlate with anchoring bias and rationalized errors [21]. Additional reasoning is productive when it widens the search across hypothesis space; it becomes destructive when it deepens commitment to an incorrect branch. Foundation models for generalist medical AI are expected to flexibly interpret combinations of imaging, records, genomics, and text while demonstrating advanced reasoning [22]; the computational view provides the framework for directing that capability where it matters most.

Reasoning as a clinical resource. The core question in advanced medical AI is not whether machines can reason in the abstract, but whether more reasoning—human, machine, or combined—changes the next real decision for a patient.

The argument has three parts. First, disease classes differ sharply in how much computation can improve outcomes. Second, patient access to longitudinal records changes the location of medical intelligence: compute can move with the patient rather than remaining trapped inside an institution. Third, the future of high-value health AI is N-of-1: iterative, longitudinal, individualized, and tied to decisions rather than one-shot answers.

Definition. A *computational view of medicine* treats disease management as the problem of converting fragmented, multimodal, longitudinal evidence into better patient-specific decisions under uncertainty.

2 Scope and evidence selection

This is a scientific perspective, not a systematic review. It supports a specific claim: that the clinical value of AI concentrates in disease settings where the bottleneck is patient-specific synthesis under uncertainty. We prioritized five evidence categories: prospective interventional studies where computation changed outcomes; mechanistic studies that make the search-space argument concrete; studies on patient data access; N-of-1 and individualized therapeutic design; and implementation work on digital phenotyping, quality, and safety [12, 15, 19].

Not all cited domains have mature evidence for the full paradigm. We distinguish three layers: direct evidence that computation changed outcomes; translational evidence that a domain is intrinsically computational; and forward-looking inference about how patient-held data could alter care models. Prospective claims should be read as extrapolations from current evidence, not settled fact.

3 A framework for computational leverage

We propose five properties that determine whether a disease class is likely to benefit from larger inference budgets:

1. **Search-space size**: how many plausible hypotheses, targets, interventions, or sequences must be evaluated.
2. **Data burden**: how many modalities and how much longitudinal context are required to reason well.
3. **Personalization**: how much the correct answer depends on the specific patient rather than the average protocol.
4. **Actionability**: whether better synthesis can materially alter diagnosis, escalation, treatment choice, or trial eligibility.
5. **Compute elasticity**: whether additional reasoning depth is likely to improve the result instead of merely restating the same answer more fluently.

Precision oncology, rare-disease genomics, resistant infection, critical care syndromes, and long-horizon chronic conditions rise to the top. They combine large search spaces with heterogeneous data and decisions whose quality directly affects outcome. Diseases governed by narrow protocols benefit more from automation than from deeper inference.

In low-elasticity domains, a clinician at normal pace already reaches the correct decision—more thinking time buys nothing. In high-elasticity domains, every additional minute of deliberation can shift the differential, reorder treatment options, or surface a trial that would otherwise be missed. Clinicians cannot spend an hour on every case, even when an hour would help. AI does not replace clinical reasoning here. It extends the reasoning budget for the cases that need it most.

The magnitude of the current reasoning gap makes this urgent. A molecular tumor board spends 15–30 minutes per case against a decision space of hundreds of drugs, thousands of trials, and millions of papers; the team searches a fraction of one percent. A geneticist reviewing a rare disease case has 60 minutes and a patient file accumulated over years; the variant literature runs to millions of entries. A primary care physician managing a multimorbid patient on twelve medications has 15 minutes; the combinatorial interaction space is enormous, navigated almost entirely from memorized heuristics. This *compute deficit*—the gap between what is searchable and what is actually searched—is where the largest clinical returns will come from.

This is the practical bridge between an abstract computational framing and the hands-on work of medical professionals. The relevant unit is not "AI reasoning" in isolation. It is the total reasoning capacity available to a clinician-patient-team before a branch is chosen. Some of that capacity comes from the clinician's expertise, some from specialist consultation, some from pre-computed evidence such as guidelines and reviews, and increasingly some from machine inference over the longitudinal record. The computational view matters because it names the bottleneck clinicians feel every day: not enough time to search deeply enough before acting.

Figure 1 makes the compute deficit concrete using epidemiological data. The scale of diagnostic error—795,000 serious harms per year in the US alone, concentrated in the same high-burden domains this paper identifies as computationally tractable—suggests that the reasoning gap is not a theoretical concern but a measurable public health problem.

The scores in Table 1 are conceptual, but they point toward a falsifiable research programme. *Search-space size* can be measured by counting plausible differentials or the relevant variant space; *compute elasticity* can

Domain	Search	Data	Pers.	Action	Elast.	Interpretation
Precision oncology	5	5	5	5	5	Dense multimodal evidence and many branching intervention paths make more inference directly decision-relevant.
Rare-disease genomics	5	5	5	4	5	Large hypothesis spaces and sparse distributed clues create unusually high value for long-context reasoning.
Resistant infection	4	4	4	5	4	The answer depends on organism, host state, prior exposure, and time-sensitive escalation.
Sepsis / ICU	4	5	4	5	4	Continuous streaming data create a high-burden monitoring problem in which timing dominates outcome.
Chronic inflammatory disease	3	4	4	4	3	Longitudinal symptom and biomarker integration can improve timing and sequencing, but action spaces are narrower.
Routine protocol care	1	2	1	2	1	Most value lies in reliable execution and logistics rather than larger inference budgets.

Table 1: Illustrative scoring matrix for computational leverage. Scores are conceptual rather than empirical, but they make the framework explicit and falsifiable.

be quantified by systematically varying reasoning depth and measuring downstream decision quality; *data burden* can be assessed by counting distinct modalities required for a defensible decision. These metrics transform the computational view from metaphor into engineering target.

Why prevalence is not enough. Common diseases are not automatically the best targets for large inference budgets. The relevant question is whether more synthesis can change a meaningful patient-specific decision. That is why prevalence rankings and compute-opportunity rankings diverge.

4 Why patient-held data matter

Clinical reasoning is not a single-agent process. It is distributed across general practitioners, specialists, pathologists, radiologists, and multidisciplinary teams. Each referral is a computational step; each handoff is a potential information loss. The rare-disease "diagnostic odyssey"—averaging 7 years and 15 referrals—is not a story about one doctor who failed to think hard enough. It is a distributed search that could not maintain state across its own nodes: hypotheses dropped at specialty boundaries, prior exclusions repeated because no entity held the full search history.

AI is uniquely positioned to hold the full search state together across fragmented care. The patient is the only entity that persists across all episodes, making them the natural locus of computational state. Once patients can assemble data across encounters, institutions, and modalities, compute moves with the patient rather than remaining trapped inside an institution. OpenNotes showed that patients value direct access to clinicians' notes and that transparency improves engagement [7]. Patients shared those notes with family and caregivers, turning records into coordination artifacts [14]. Patient-generated data—symptom tracking, wearables, home monitoring—are increasingly common [25]. Records are no longer only institutional memory; they are becoming a substrate for continuous reasoning.

Raw access is not enough. Precision medicine remains incomplete without patients as active participants

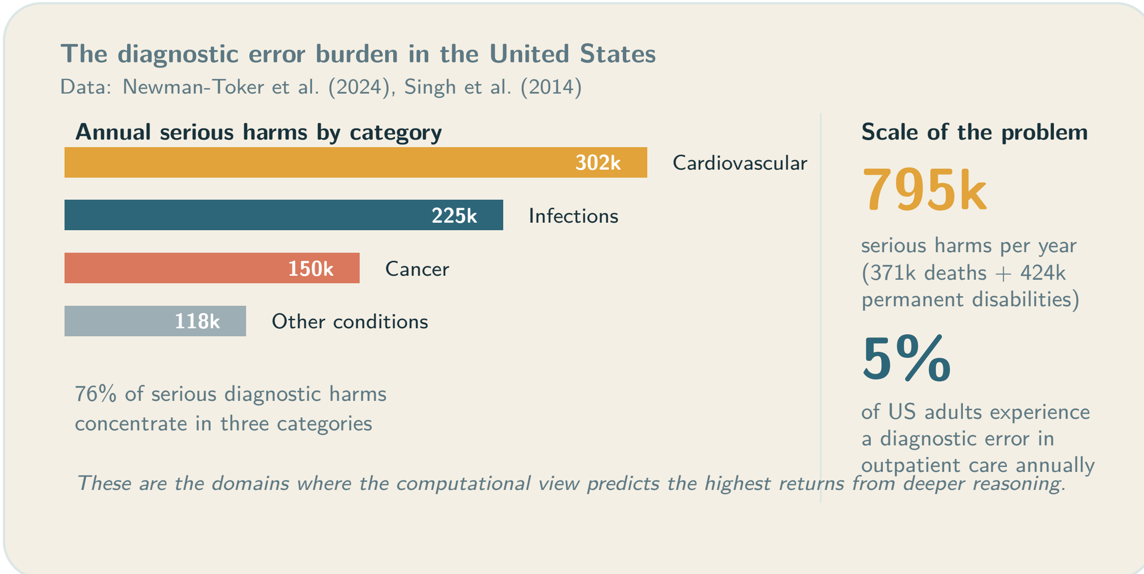

Figure 1: The diagnostic error burden concentrates in exactly the domains where the computational view predicts the highest compute elasticity. Data from Newman-Toker et al. [24] and Singh et al. [31].

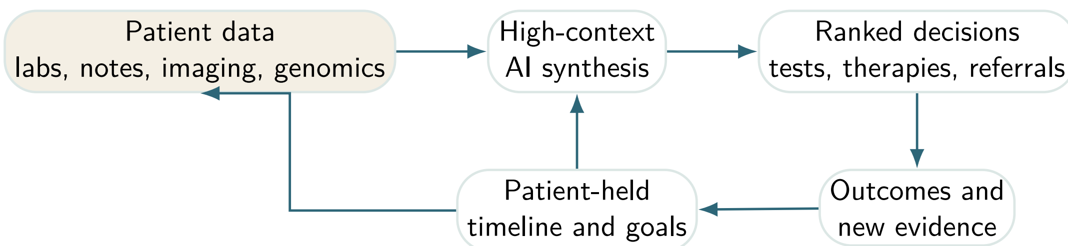

Figure 2: The patient-compute loop. In high-burden disease, value comes from repeated synthesis over the full timeline rather than one-shot question answering.

in data interpretation [35]. AI for shared decision-making must improve deliberation, not simply generate recommendations [11]. Integration of patient-generated health data is not merely a technical challenge but a clinical, organizational, and evidentiary one [15]. Early work shows patients can already use large language models on open notes to interpret their records [29]—early evidence, not clinical validation, but it clarifies the direction. Once patients can access their full timeline, the missing layer is repeatable reasoning over it.

4.1 Counterarguments and constraints

Patient-held data are not automatically useful. Provenance may be uncertain, timestamps incomplete, self-tracked measures drifting, and the patients most likely to benefit may face the largest barriers in digital access [12]. The computational paradigm depends on a stricter standard than interoperability: provenance, chronology, validation, and interfaces that separate observed evidence from inferred conclusions.

Access versus leverage. Data access creates possibility. Computational interpretation creates leverage. In a patient-centered health stack, the two should be designed together.

5 Evidence from the highest-leverage domains

5.1 Precision oncology

Cancer is already a computational problem. Molecular diagnosis, resistance interpretation, treatment sequencing, and trial matching require synthesis across pathology, imaging, genomics, prior therapy, and time. Personalized cancer vaccines make this explicit: identifying patient-specific tumor mutations, ranking neoantigens, and manufacturing a bespoke intervention. Sahin and colleagues showed that individualized RNA mutanome vaccines mobilize poly-specific immunity against melanoma [28]. AlphaFold expands the search surface further by making protein consequence analysis tractable across large molecular spaces [17].

Compute sits inside the decision itself: which mutation matters, which escape mechanisms are plausible, which combination is worth pursuing, which trial is defensible for this patient now. The burden of reasoning has become part of the disease burden.

A single patient may generate surgical pathology, radiology progression patterns, panel sequencing, circulating tumor DNA, treatment toxicities, and eligibility-relevant comorbidities. The computational task is to rank drivers, resistance hypotheses, and therapeutic branches while making uncertainty visible. The leverage comes not from generating prose about cancer but from compressing a large search problem into a smaller set of defensible next moves.

5.2 Rare disease genomics

The problem is not only finding a variant but aligning phenotype, pedigree, prior negative workup, and the literature fast enough to affect care. Wojcik and colleagues showed in the *New England Journal of Medicine* that genome sequencing materially improved diagnosis for suspected rare disease [36]. This is a canonical compute-heavy domain: large hypothesis spaces, sparse signal, fragmented records, and a direct payoff when reasoning succeeds. The cost of failure is now quantified: rare-disease patients accumulate average healthcare costs of €27,000 over their diagnostic odyssey, compared to €3,600 for matched controls, and avoidable costs of delayed diagnosis in the US range from $86,000 to over $517,000 per patient [8].

5.3 Resistant infection and antibiotic discovery

Antimicrobial resistance presents a two-layer search problem. The first is patient-specific: organism identity, susceptibility, host state, prior exposure, organ function, and drug interactions. The second is discovery: searching chemical space for compounds with novel activity. Stokes and colleagues used deep learning to identify halicin, accelerating antibiotic discovery in regions of chemical space invisible to standard pipelines [33]. More compute means better regimen ranking in the clinic and better exploration of candidate space in the lab.

5.4 Sepsis and critical care

In critical care the challenge is time. Dozens of measurements update continuously, and the relevant signal is distributed across trends rather than single observations. A machine-learning sepsis early-warning system deployed across multiple hospitals was associated with faster antibiotic administration and lower mortality [1]. Deep learning on electronic health records can scale across broad clinical prediction tasks using the raw longitudinal record [27], and pragmatic deterioration surveillance can attach to live escalation pathways

[23]. These results establish that longitudinal records contain actionable signal, and systems that hold more context together improve timing-sensitive decisions.

5.5 Patient-reported outcomes and symptom monitoring

Not all high-value computation depends on genomics. Electronic patient-reported symptom monitoring during cancer treatment improved overall survival, likely by enabling earlier response to deterioration patterns otherwise missed [3, 4]. Symptoms, side effects, and tolerability live outside the hospital until severe enough to trigger contact. In the computational paradigm, those signals become first-class inputs into the case model.

6 N-of-1 medicine and personal computation

The computational view converges with N-of-1 medicine. Schork argued that precision medicine must be understood at the level of the individual, not the cohort average [30]. Guyatt laid the foundation for randomized trials in individual patients [10]; Lillie argued that N-of-1 trials align with the actual endpoint of clinical practice: optimizing care for one person [19]. Augustine proposed a roadmap for N-of-1 therapies covering scientific, operational, and regulatory requirements [2]; Jonker outlined how N-of-1 medicine could reshape treatment evaluation for rare and heterogeneous disease [16]. These contributions move personalization from slogan to organizing scientific unit.

Longitudinal self-measurement provides parallel evidence. Integrative personal omics profiling reveals dynamic molecular phenotypes within a single individual over time [5]. Dense personal data clouds can monitor health transitions with temporal resolution far exceeding episodic care [26]. Biology, symptoms, and risk are often more legible in longitudinal within-person data than in sparse snapshots.

Public self-experimentation—highly instrumented programmes such as Bryan Johnson's—is not clinical evidence. What it reveals is latent demand: people want to run dense longitudinal data loops over their own bodies. The scientific path forward is audited N-of-1 workflows combining patient goals, validated measurements, and clinically accountable reasoning.

7 Case examples

Table 2 shows why a single computational model fits these otherwise different domains.

7.1 Concrete pathways

In each case, computation improves preparation, ranking, and timing for the next branch in care—not to replace the clinician but to strengthen the decision.

1. **Precision oncology.** A patient-assembled case file combines sequencing, pathology, imaging, and prior therapies. Compute ranks drivers, resistance mechanisms, toxic trade-offs, and live trials. The output is a defensible next branch with explicit uncertainty.
2. **Rare disease.** A phenotype timeline with milestones, regressions, imaging, and prior exclusions is linked to genome evidence and literature, eliminating time spent re-deriving the case at each referral.

Domain	Core data	How computation is leveraged	Representative evidence
Precision oncology	Genomics, pathology, imaging, treatment history	Variant interpretation, resistance modeling, neoantigen ranking, regimen sequencing, trial matching	Sahin et al. [28]; Jumper et al. [17]
Rare disease	Phenotype timeline, pedigree, genome sequencing, prior workup	Cross-linking phenotype ontologies, literature, family history, and candidate variants	Wojcik et al. [36]
Resistant infection	Microbiology, susceptibilities, organ function, prior antibiotic exposure	Better ranking of therapies and faster search across compound or regimen space	Stokes et al. [33]
Sepsis / ICU	Streaming vitals, labs, notes, medication events	Earlier escalation from weak, time-distributed patterns across the longitudinal record	Adams et al. [1]; Rajkomar et al. [27]; Moses et al. [23]
Patient-reported symptoms	Patient-reported outcomes, treatments, toxicities, goals	Detecting deterioration and treatment intolerance before they become acute	Basch et al. [3]; Basch et al. [4]
Patient-held records	Notes, meds, labs, imaging reports, symptoms, goals	Converts raw data access into usable reasoning leverage before and between encounters	Delbanco et al. [7]; Salmi et al. [29]

Table 2: High-value disease areas under a computational lens.

3. **Chronic inflammatory disease.** Symptoms, labs, wearable data, and medication exposure are integrated within-person to identify precursors of flare, intolerance, or non-response.

8 Evaluation and endpoints

The computational view should change how health AI is evaluated. Discrimination metrics are not enough. In high-stakes medicine, calibration matters as much as accuracy: a system that outputs "60% diagnosis A, 25% diagnosis B, and here is the test that discriminates" is more useful than a bare point estimate, even at identical accuracy. More reasoning sometimes does not change the top-ranked answer but sharpens the uncertainty estimate, which clinically determines whether a clinician orders a confirmatory test, refers, or treats empirically. The relevant question is whether synthesis quality—including calibration—improved at the point where the case could still change direction.

9 What should be built

The practical implication is narrower than much current health-AI rhetoric. The field does not primarily need more conversational surfaces. It needs reasoning infrastructure: systems that assemble patient records

Future patient personas

Patient	Data substrate	Computational leverage
Asha, 58 metastatic NSCLC	Tumour DNA/RNA, pathology, CT trends, prior lines, toxicities, performance status	Variant prioritization, resistance hypotheses, neoantigen ranking, and live trial matching change the next therapeutic branch.
Mateo, 6 undiagnosed neuro-developmental syndrome	Phenotype timeline, EEG, MRI, family history, genome sequencing, prior negative workup	Phenotype-variant ranking and literature synthesis narrow candidate mechanisms and guide targeted confirmatory testing.
Leila, 34 Crohn's disease on biologics	Symptoms, calprotectin, medication adherence, wearables, sleep, diet, clinician notes	Within-person pattern detection turns fragmented self-tracking into actionable flare prediction and better-timed escalation.

Figure 3: Concrete personas for a computational health stack. The same design principle appears across specialties: assemble the full timeline, compute over it repeatedly, and use the result to improve the next decision.

across sites, preserve chronology, separate evidence from inference, and return ranked next steps rather than fluent summaries. The right unit is a living case model that can be re-run as disease evolves.

These systems should strengthen the clinical encounter rather than bypass it. The design target is a patient-clinician conversation with better state, better uncertainty handling, and better preparation before the next branch in care is chosen.

10 Risks and constraints

The computational view sharpens the need for evidence and oversight. High-burden disease is exactly where hallucinated synthesis, hidden bias, and brittle prompting are most dangerous. Reasoning systems must therefore be auditable, source-grounded, and explicit about what they do not know.

Not every disease is compute-elastic. Some problems remain dominated by biology, logistics, or protocol execution rather than reasoning depth. And if compute meaningfully improves outcomes, unequal access to compute becomes a health inequity. A health stack that gives patients records without usable synthesis leaves too much leverage with the institution.

11 Operational consequences

The agenda for health AI changes. Clinical systems should separate protocol problems from search problems. Regulators should ask not only whether a model is accurate on average, but whether it improves reasoning quality in the cases where reasoning is the bottleneck. And product teams should direct the largest inference budgets toward cases where better synthesis can change the next branch in care.

Study type	What changed	Why it matters for the thesis
Pragmatic randomized clinical trial in cardiovascular screening	AI-enabled ECG alerts were tested as an intervention rather than a retrospective classifier [20]	Supports the claim that computational systems can change downstream decisions and outcomes, not merely generate scores.
Pragmatic cluster-randomized deterioration trial	Real-time surveillance for patient deterioration was tested across live inpatient settings [23]	Strengthens the case that high-context clinical surveillance can function as an operational care system.
Patient-record sharing study	Patients with transparent notes shared them with family and caregivers, extending the practical reach of the record [14]	Shows that once patients have access to records, those records begin acting as coordination infrastructure.
Patient-side LLM proof of concept	Patients used large language models on open notes to better understand and work with their records [29]	Gives direct support to the argument that personal compute can become part of the patient toolkit.

Table 3: Evidence that the computational view extends beyond retrospective prediction.

The practical payoff is concentration: fewer missed driver mutations, faster rare-disease diagnoses, better antibiotic choices, earlier sepsis escalation, and better continuity across fragmented care. Pragmatic intervention studies in ECG alerting, deterioration surveillance, and symptom monitoring suggest that this shift can move from theory to operations when attached to real decision paths [3, 20, 23].

12 Conclusion

Medicine is already a computational process. Evidence-based medicine formalized that process at the population level. The computational view proposed here formalizes it at the patient level and argues that a large share of avoidable harm in complex disease comes from the gap between what could be searched, synthesized, and updated for a patient and what is actually computed in routine care.

The diseases that matter most under this view are not simply the most prevalent. They are the ones that become search problems at patient resolution: precision oncology, rare-disease genomics, resistant infection, selected critical care settings, and chronic relapsing disease. In these domains, better reasoning can change the next branch in care.

The deeper implication is structural. If reasoning has historically been scarce and expensive—bound to the time, memory, and attention of individual clinicians—then cheap, portable computation changes not only the quality of individual decisions but the architecture of medicine itself. A computational view of medicine therefore does two things at once: it shows where additional inference can genuinely improve outcomes, and it points toward a health system in which understanding becomes a shared, longitudinal, patient-held capability rather than a privilege of the institution.

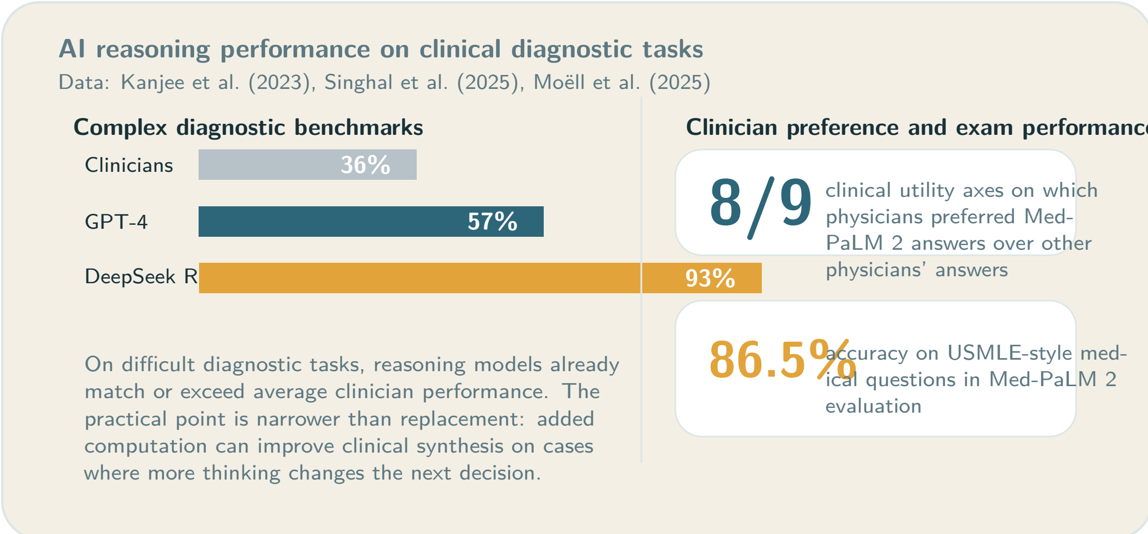

Figure 4: Reasoning-focused AI systems already perform strongly on difficult diagnostic and medical benchmarks. Data from Kanjee et al. [18], Singhal et al. [32], and Moëll et al. [21]. The clinical relevance is practical: in some cases, more computation improves the quality of synthesis.

References

[1] Roy Adams, Katharine E. Henry, Anirudh Sridharan, Hossein Soleimani, Andong Zhan, Nishi Rawat, Lauren Johnson, David N. Hager, Sara E. Cosgrove, Andrew Markowski, et al. Prospective, multi-site study of patient outcomes after implementation of the TREWS machine learning-based early warning system for sepsis. *Nature Medicine*, 28(7):1455–1460, 2022. doi: 10.1038/s41591-022-01894-0.

[2] Emily F. Augustine, Melissa Chiong, Ans de Wilde, Eran Elinav, David C. Fajgenbaum, Terence R. Flotte, David B. Goldstein, Rachelle C. Green, Melissa A. Haendel, Lawrence T. Mazzola, et al. A roadmap for n-of-1 therapies. *Nature Reviews Drug Discovery*, 23(7):507–509, 2024. doi: 10.1038/d41573-024-00087-3.

[3] Ethan Basch, Amy M. Deal, Mark G. Kris, Howard I. Scher, Clifford A. Hudis, Paulina Sabbatini, Lauren Rogak, Ann V. Bennett, Allison C. Dueck, Timothy M. Atkinson, et al. Symptom monitoring with patient-reported outcomes during routine cancer treatment: A randomized controlled trial. *Journal of Clinical Oncology*, 34(6):557–565, 2016. doi: 10.1200/JCO.2015.63.0830.

[4] Ethan Basch, Deborah Schrag, Sara Henson, Jan Jansen, Benjamin Ginos, Ann Stover, Paula Carr, Patricia Spears, Christopher Jonas, Melissa Weiser, et al. Effect of electronic symptom monitoring on patient-reported outcomes among patients with metastatic cancer: A randomized clinical trial. *JAMA*, 318(2):197–198, 2017. doi: 10.1001/jama.2017.7156.

[5] Rui Chen, George I. Mias, Jennifer Li-Pook-Than, Long Jiang, Hugo Y. K. Lam, Rui Chen, Eyal Miriami, Konrad J. Karczewski, Muthu Hariharan, Frederick E. Dewey, et al. Personal omics profiling reveals dynamic molecular and medical phenotypes. *Cell*, 148(6):1293–1307, 2012. doi: 10.1016/j.cell.2012.02.009.

[6] Pat Croskerry. A universal model of diagnostic reasoning. *Academic Medicine*, 84(8):1022–1028, 2009. doi: 10.1097/ACM.0b013e3181ace703.

[7] Tom Delbanco, Jan Walker, Jonathan D. Darer, Joann G. Elmore, Henry J. Feldman, Suzanne G. Leveille, James D. Ralston, Stephen E. Ross, Elisabeth Vodicka, and Valerie D. Weber. Open

Domain	Primary endpoint candidates	Why these endpoints fit the computational thesis
Rare disease	Time to diagnosis; diagnostic yield; avoided procedures	Measures whether synthesis reduced time spent on the wrong branch of care.
Precision oncology	Time to molecular recommendation; trial-match yield; change in treatment ranking; time to next line	Measures whether computation improved branching decisions rather than documentation alone.
Sepsis / ICU	Time to escalation; antibiotics or intervention timing; false alerts per patient-day; mortality or ICU transfer	Measures whether longitudinal synthesis improved timing without overwhelming workflow.
Chronic disease / N-of-1 care	Time to flare detection; within-person treatment response; symptom burden; adherence to preferred regimen	Measures whether repeated computation improves individualized management over time.
Patient-facing record synthesis	Patient comprehension; visit preparedness; question quality; fewer duplicated tests; continuity across sites	Measures whether access plus reasoning improved the practical utility of records.

Table 4: Evaluation endpoints aligned with a computational view of medicine. The goal is to measure whether reasoning improves decisions, not only whether predictions look accurate offline.

notes: Doctors and patients signing on. *Annals of Internal Medicine*, 153(2):121–125, 2010. doi: 10.7326/0003-4819-153-2-201007200-00008.

[8] Rick Glaubitz, Luise Heinrich, Falko Tesch, Martin Seifert, Katrin Christiane Reber, Ursula Marschall, Jochen Schmitt, and Gabriele Müller. The cost of the diagnostic odyssey of patients with suspected rare diseases. *Orphanet Journal of Rare Diseases*, 20:222, 2025. doi: 10.1186/s13023-025-03751-y.

[9] Mark L. Graber, Nancy Franklin, and Ruthanna Gordon. Diagnostic error in internal medicine. *Archives of Internal Medicine*, 165(13):1493–1499, 2005. doi: 10.1001/archinte.165.13.1493.

[10] Gordon Guyatt, David Sackett, Jonathan Adachi, Roger Roberts, Jose Chong, David Rosenbloom, and Joseph Keller. A clinician's guide for conducting randomized trials in individual patients. *Canadian Medical Association Journal*, 134(8):889–892, 1986.

[11] Iva Hassan, Alastair van Heerden, Bala Padmanabhan, Norbert C. Benda, Emanuele Mazzone, Leo Anthony Celi, and Claudia C. Dobler. How can artificial intelligence support shared decision making? a systematic review of ai-enabled decision aids. *npj Digital Medicine*, 7(1):46, 2024. doi: 10.1038/s41746-024-01056-1.

[12] Kit Huckvale, Seema Venkatesh, and Helen Christensen. Toward clinical digital phenotyping: a timely opportunity to consider purpose, quality, and safety. *npj Digital Medicine*, 2(1):88, 2019. doi: 10.1038/s41746-019-0166-1.

[13] Greg Irving, Ana Luisa Neves, Hajira Dambha-Miller, Ai Oishi, Hiroko Tagashira, Anistassia Verho,

and John Holden. International variations in primary care physician consultation time: a systematic review of 67 countries. *BMJ Open*, 7(10):e017902, 2017. doi: 10.1136/bmjopen-2017-017902.

[14] Sara L. Jackson, Roanne Mejilla, Jonathan D. Darer, Natalia V. Oster, James D. Ralston, Suzanne G. Leveille, Jan Walker, and Tom Delbanco. Patients who share transparent visit notes with others: characteristics, risks, and benefits. *Journal of Medical Internet Research*, 16(11):e247, 2014. doi: 10.2196/jmir.3363.

[15] Heather S. L. Jim, Andrea I. Hoogland, Noah C. Brownstein, Andre Barata, Adam P. Dicker, Hans Knoop, Brenda D. Gonzalez, Robert Perkins, Dana Rollison, Rita Nanda, et al. Innovations in research and clinical care using patient-generated health data. *CA: A Cancer Journal for Clinicians*, 70(3): 182–199, 2020. doi: 10.3322/caac.21608.

[16] Aafke H. Jonker, Costas Demetzos, Anthonius de Boer, Eric G. E. de Vries, Johan T. den Dunnen, Marc T. W. Handley, Anke W. Hoes, Hubert G. M. Leufkens, Jean J. Sikkens, Rosalie van der Graaf, et al. N-of-1 medicine and treatment outcomes: the future is bright. *Nature Reviews Drug Discovery*, 24(5):335–336, 2025. doi: 10.1038/d41573-025-00004-z.

[17] John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873):583–589, 2021. doi: 10.1038/s41586-021-03819-2.

[18] Zahir Kanjee, Byron Crowe, and Adam Rodman. Accuracy of a generative artificial intelligence model in a complex diagnostic challenge. *JAMA*, 330(1):78–80, 2023. doi: 10.1001/jama.2023.8288.

[19] Elizabeth O. Lillie, Bence Patay, Judith Diamant, Brian Issell, Eric J. Topol, and Nicholas J. Schork. The n-of-1 clinical trial: the ultimate strategy for individualizing medicine? *Personalized Medicine*, 8 (2):161–173, 2011. doi: 10.2217/pme.11.7.

[20] Chin-Sheng Lin, Wei-Ting Liu, Dung-Jang Tsai, Yu-Sheng Lou, Chiao-Hsiang Chang, Chiao-Chin Lee, Wen-Hui Fang, Chih-Chia Wang, Yen-Yuan Chen, Wei-Shiang Lin, et al. Ai-enabled electrocardiography alert intervention and all-cause mortality: a pragmatic randomized clinical trial. *Nature Medicine*, 30 (5):1461–1470, 2024. doi: 10.1038/s41591-024-02961-4.

[21] Birger Moëll, Fredrik Sand Aronsson, and Sanian Akbar. Medical reasoning in LLMs: an in-depth analysis of DeepSeek R1. *Frontiers in Artificial Intelligence*, 8:1616145, 2025. doi: 10.3389/frai.2025.1616145.

[22] Michael Moor, Oishi Banerjee, Zahra Shakeri Hossein Abad, Harlan M. Krumholz, Jure Leskovec, Eric J. Topol, and Pranav Rajpurkar. Foundation models for generalist medical artificial intelligence. *Nature*, 616(7956):259–265, 2023. doi: 10.1038/s41586-023-05881-4.

[23] Rebecca Moses, Lu Chen, Irtiza Khan, Zachary Pruitt, Andrew Sharp, Yu Ruan, Yiyun He, Jason Stifter, Hossein Estiri, Stephen A. Collins, et al. Real-time surveillance system for patient deterioration: a pragmatic cluster-randomized controlled trial. *Nature Medicine*, 2025. doi: 10.1038/s41591-025-03609-7.

[24] David E. Newman-Toker, Najlla Nassery, Adam C. Schaffer, Chihwen Winnie Yu-Moe, Gwendolyn D. Clemens, Zheyu Wang, Yuxin Zhu, Ali S. Saber Tehrani, Mehdi Fanai, Ahmed Hassoon, and Dana Siegal. Burden of serious harms from diagnostic error in the USA. *BMJ Quality & Safety*, 33(2): 109–120, 2024. doi: 10.1136/bmjqs-2021-014130.

[25] Linda G. Park, Felicity Ng, Jimmy Shim, Adham Elnaggar, Oscar Villero, Frank Perna, Steven M. Paul, and Christopher R. Friese. Perceptions and experiences of using mobile technology for symptom

management in patients with cancer: a systematic review. *JMIR mHealth and uHealth*, 6(1):e30, 2018. doi: 10.2196/mhealth.8748.

[26] Brian D. Piening, Wenyu Zhou, Kevin Contrepois, Hannes Röst, Gabriela J. Gu Urban, Tanzeem Mishra, Brett M. Hanson, Elizabeth J. Bautista, Steven R. Leopold, Chih-Ying Yeh, et al. Integrative personal omics profiles during periods of weight gain and loss. *Cell Systems*, 6(2):157–170.e8, 2018. doi: 10.1016/j.cels.2018.01.013.

[27] Alvin Rajkomar, Eyal Oren, Kai Chen, Andrew M. Dai, Nissan Hajaj, Michaela Hardt, Peter J. Liu, Xiaobing Liu, Jake Marcus, Mimi Sun, et al. Scalable and accurate deep learning with electronic health records. *npj Digital Medicine*, 1(1):18, 2018. doi: 10.1038/s41746-018-0029-1.

[28] Ugur Sahin, Evelyna Derhovanessian, Michael Miller, Ben Kloke, Peter Simon, Markus Löwer, Thomas Bukur, Anat Tadmor, Ulrike Luxemburger, Benedikt Schrörs, et al. Personalized rna mutanome vaccines mobilize poly-specific therapeutic immunity against cancer. *Nature*, 547(7662):222–226, 2017. doi: 10.1038/nature23003.

[29] Liz Salmi, Dana M. Lewis, Jennifer L. Clarke, Zhiyong Dong, Rudy Fischmann, Emily I. McIntosh, Chethan R. Sarabu, and Catherine M. DesRoches. A proof-of-concept study for patient use of open notes with large language models. *JAMIA Open*, 8(2), 2025. doi: 10.1093/jamiaopen/ooaf021.

[30] Nicholas J. Schork. Personalized medicine: Time for one-person trials. *Nature*, 520(7549):609–611, 2015. doi: 10.1038/520609a.

[31] Hardeep Singh, Ashley N. D. Meyer, and Eric J. Thomas. The frequency of diagnostic errors in outpatient care: estimations from three large observational studies involving US adult populations. *BMJ Quality & Safety*, 23(9):727–731, 2014. doi: 10.1136/bmjqs-2013-002627.

[32] Karan Singhal, Tao Tu, Juraj Gottweis, Rory Sayres, Ellery Wulczyn, Le Hou, Kevin Clark, Stephen Pfohl, Heather Cole-Lewis, Darlene Neal, et al. Toward expert-level medical question answering with large language models. *Nature Medicine*, 31:943–950, 2025. doi: 10.1038/s41591-024-03423-7.

[33] Jonathan M. Stokes, Kevin Yang, Kyle Swanson, Wengong Jin, Andrés Cubillos-Ruiz, Nicole M. Donghia, Craig R. MacNair, Shawn French, Lindsey A. Carfrae, Zoe Bloom-Ackermann, et al. A deep learning approach to antibiotic discovery. *Cell*, 180(4):688–702.e13, 2020. doi: 10.1016/j.cell.2020.01.021.

[34] Eric J. Topol. High-performance medicine: the convergence of human and artificial intelligence. *Nature Medicine*, 25(1):44–56, 2019. doi: 10.1038/s41591-018-0300-7.

[35] Katherine R. Tuttle, Radica Z. Alicic, O. Kenrik Duru, C. Raymond Jones, Karen B. Daratha, Stephanie B. Nicholas, and Joshua J. Neumiller. Clinical characteristics of and risk factors for chronic kidney disease among adults and perspectives on patient-centered precision medicine. *Kidney International*, 99(1):52–66, 2021. doi: 10.1016/j.kint.2020.10.046.

[36] Monica H. Wojcik, Gabrielle Lemire, Eva Berger, Maha S. Zaki, Mariel Wissmann, Wathone Win, Susan M. White, Ben Weisburd, Dagmar Wieczorek, Leigh B. Waddell, et al. Genome sequencing for diagnosing rare diseases. *The New England Journal of Medicine*, 390(21):1985–1997, 2024. doi: 10.1056/NEJMoa2314761.